

AI Application Stack Overview

A TLDR Intro to AI CyberInfrastructure

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https://github.com/mcgheee/AI_02H_Course

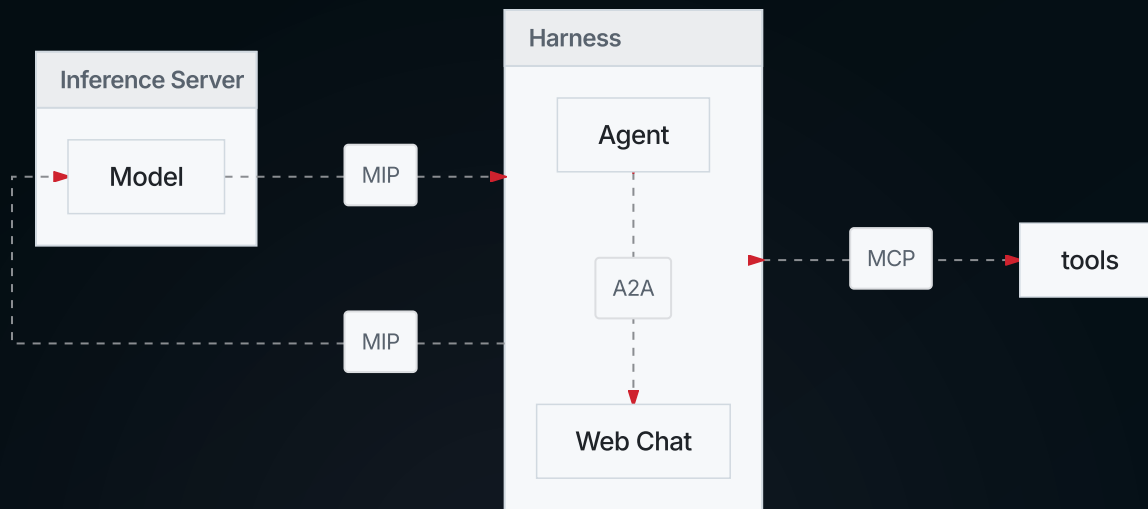
Terms

- Harness - The application that allows users or other software to interact with AI models.
- Agent - A special type of harness that operates on a loop.
- Context - What is stored in the model's working memory.
- Memory - Durable long term storage of data
- Prompt - A single instruction or question to the model.
- Turn - A single response in a conversation from either the client or the model.
- Inference - The process of generating a response from the model.

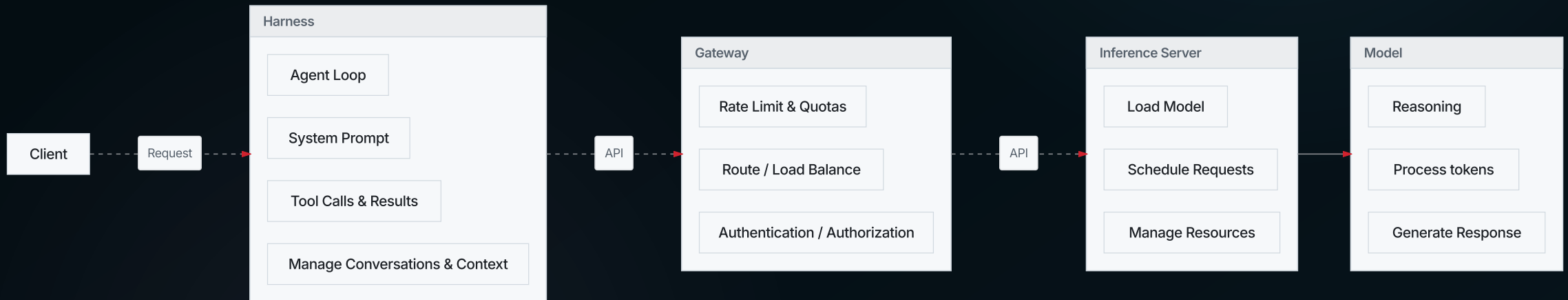
Model Types

- LLMs
 - Ex: GPT 6, Claude 5, Grok 4
- Diffusion
 - Ex: Stable Diffusion, DALL-E, LTX
- Classification
 - Ex: Jev, Laya

AI Stack Simplified



Application Workflow



Harness

- Basically any app used to interact with AI
 - Web Chat
 - CLI
 - Agentic

Inference Server

- Loads and serves the model
- Schedules requests and manages batching
- Manages CPU/GPU/accelerator resources

Agents

- A type of Harness
- Operates on a loop until conditions are met
- May provide or connect to tools

Examples: Openclaw, Hermes, OpenCode, Codex, Claude Code

Tools

- May be provided by the harness
 - Ex: Read, Write, Execute
- Or, connected to an external service via MCP
- Ex: Web scraping, Memory, Computer Use

Skills

- Provide instructions, **not** capabilities
- Written in natural language
- Usually structured in Markdown

Context

- Model's working memory
- Limited in Size
- Lossy due to compaction

Guardrails

- Constrains what a model can accept, produce, or do
- Soft constraints that guide the model's behavior
 - System / Developer Instructions, Training, Prompt
- Hard constraints that prevent action
 - Sandboxing, Permission Gating, Human Approval

